

ASTE 583 Final Project

Introduction

The ASTE 583 Final Project focuses on Lunar Trailblazer, a satellite which lost contact with earth shortly after launch in February, 2025. **Insert more detailed problem statement here.**

All scripts, functions, reports and other deliverables were written by Varsha Krishnakumar and Nicholas Mellanby.

Section 1: Problem Initialization

```
clear
clc

% Call Project Init Function
[params,sc,st,X0,P0] = projectInit();

% Init Times for Initial Detection, LTM, and LCM
[ET0,LTM,LCM] = initTime();
```

Section 2: Integrate Dynamics (Sun-Centered EMO2000)

```
%Set Options
options = odeset('RelTol',1e-10,'AbsTol',1e-10);
step = 1800; %30 min intervals

%Set X0 pre LTM
X0preLTM = [X0;reshape(eye(10),100,1)];
X0preLTM(7) = 0.6996824;

%Integrate to LTM
[TpreLTM,XpreLTM] =
ode45(@projectDynamics,ET0:step:LTM,X0preLTM,options,params,sc);

%Set X0 post LTM
X0postLTM = XpreLTM(end,:);

%Apply deltaV
deltaVLTM = [0.6931075; -0.8462091; 0.0956979];
X0postLTM(4:6) = X0postLTM(4:6) + deltaVLTM;

%Integrate to LCM
```

```

[TpostLTM,XpostLTM] =
ode45(@projectDynamics,LTM:step:LCM,X0postLTM,options,params,sc);

%Set X0 post LCM
X0postLCM = XpostLTM(end,:); %Don't apply to reference trajectory

%Integrate from LCM to 251 days – LCM occurs after 24 days, so add 227 days
to trajectory
TOF = 227*24*3600; %236 days in seconds
[TpostLCM,XpostLCM] =
ode45(@projectDynamics,LCM:step:LCM+TOF,X0postLCM,options,params,sc);

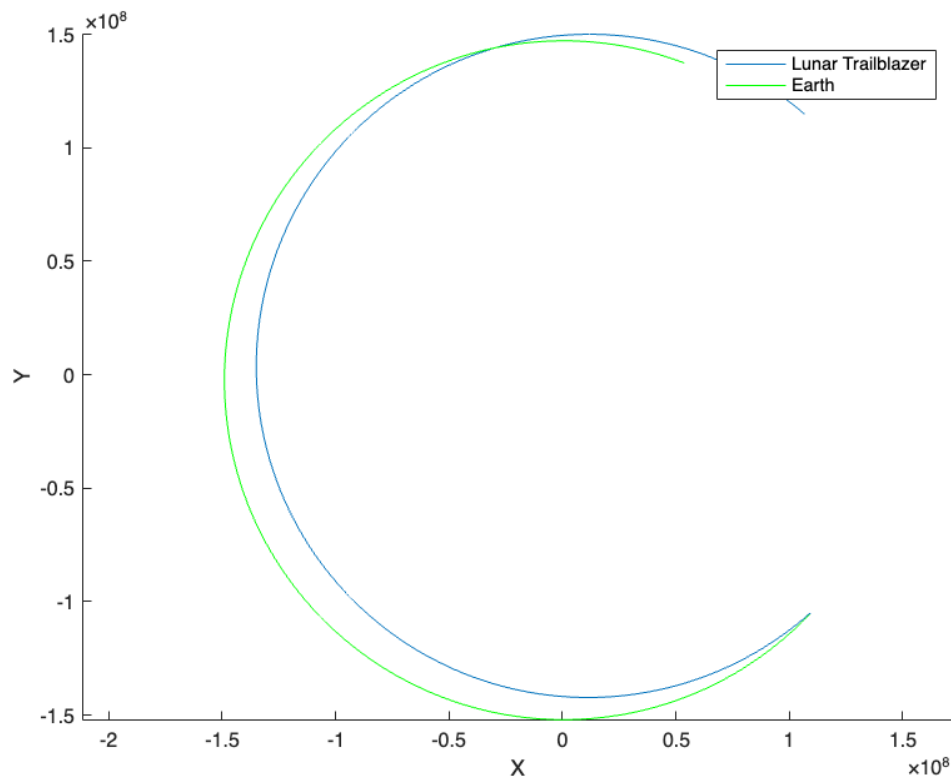
%Combine Results
Xtotal = [XpreLTM(:,:);XpostLTM(2:end,:);XpostLCM(2:end,:)]; %omit the
first element to elim overlap
Ttotal = [TpreLTM;TpostLTM(2:end);TpostLCM(2:end)];

%Plot Sun-Centric Trajectory
figure();
hold on
plot3(Xtotal(:,1),Xtotal(:,2),Xtotal(:,3))
axis('equal')
xlabel("X")
ylabel("Y")

%Retrieve Earth Position at Each Time Step
xEarth = cspice_spkezr('Earth',Ttotal(:,1)', 'ECLIPJ2000', 'NONE','Sun');
plot3(xEarth(1,:),xEarth(2,:),xEarth(3,:), '-g')

%Add Legend
legend("Lunar Trailblazer","Earth")

```

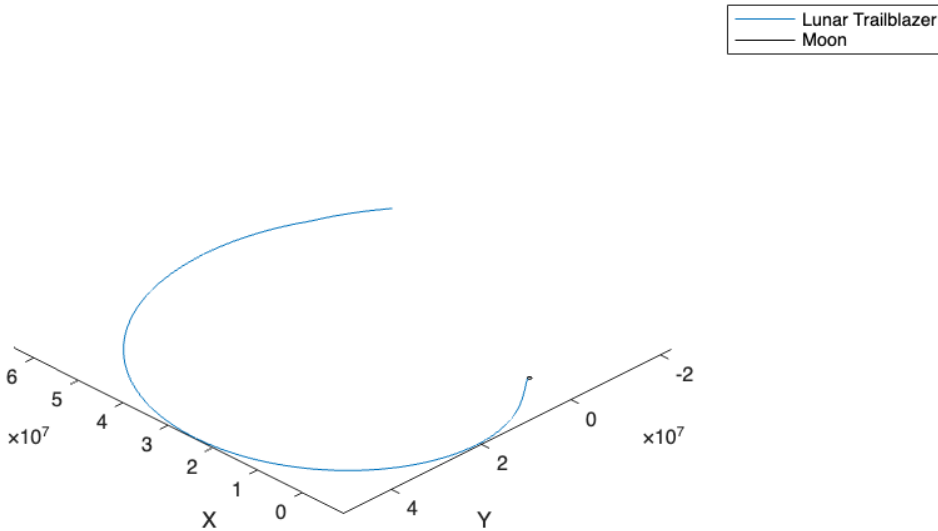


Section 3: Convert Plot to Earth Centered EMO2000

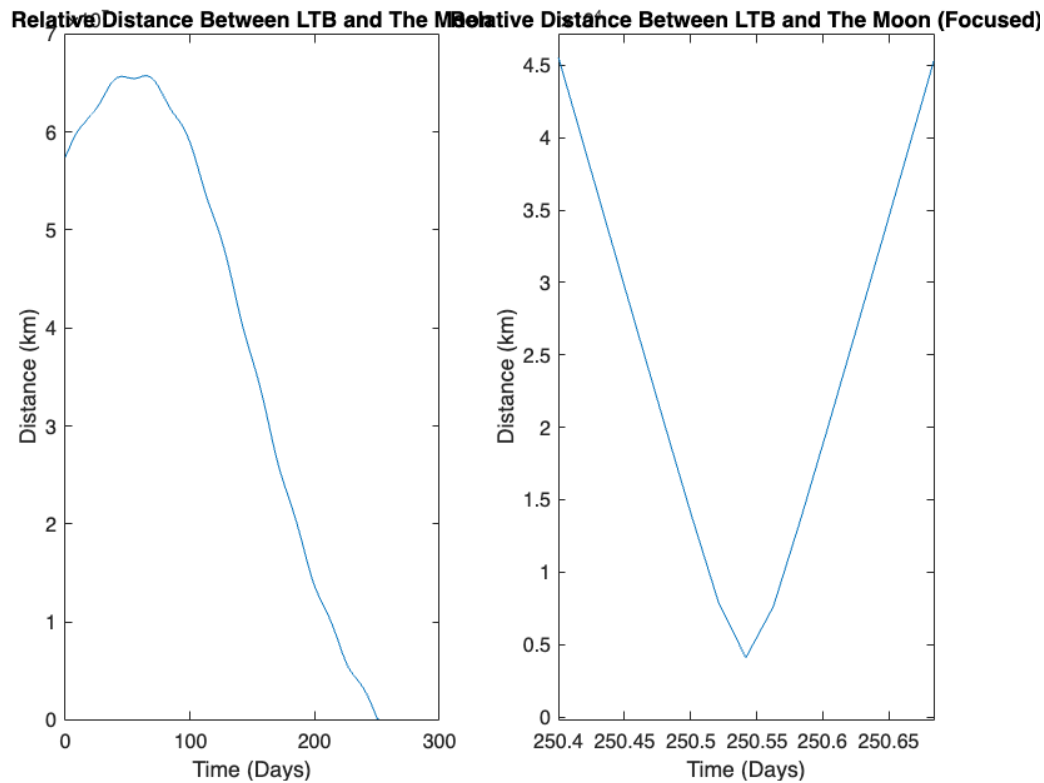
```
% Plot Relative Earth Motion
XrelEarth = Xtotal(:,1:6) - xEarth'; %Ref Traj
%XrelEarth = Xsamp(:,1:6) - xEarth'; %Samp Traj
figure();
hold on
plot3(XrelEarth(:,1),XrelEarth(:,2),XrelEarth(:,3))
view([-135 30])

%Add Moon
%Retrieve Earth Position at Each Time Step
xMoon = cspice_spkezr('Moon',Ttotal(:,1)', 'ECLIPJ2000', 'NONE','Earth');
plot3(xMoon(1,:),xMoon(2,:),xMoon(3,:), '-k')

axis("equal")
xlabel("X")
ylabel("Y")
legend("Lunar Trailblazer","Moon")
```



```
%Verify Lunar Flyby
flybyDistance = zeros(length(XrelEarth),1);
minflybyDist = 10^9; %Unrealistic flyby distance number
timeminflybyDist = 0; %Unrealistic flyby time
for i = 1:length(XrelEarth)
    flybyDistance(i,1) = norm(XrelEarth(i,1:3) - xMoon(1:3,i)');
    if(flybyDistance(i,1) < minflybyDist)
        minflybyDist = flybyDistance(i,1);
        timeminflybyDist = (Ttotal(i)-ET0)/(24*3600); %Time since ET0 in
days
    end
end
figure();
subplot(1,2,1)
plot((Ttotal-ET0)/(24*3600),flybyDistance)
title("Relative Distance Between LTB and The Moon")
xlabel("Time (Days)")
ylabel("Distance (km)")
subplot(1,2,2)
plot((Ttotal-ET0)/(24*3600),flybyDistance)
xlim([250.400 250.684])
ylim([-207 47165])
title("Relative Distance Between LTB and The Moon (Focused)")
xlabel("Time (Days)")
ylabel("Distance (km)")
```



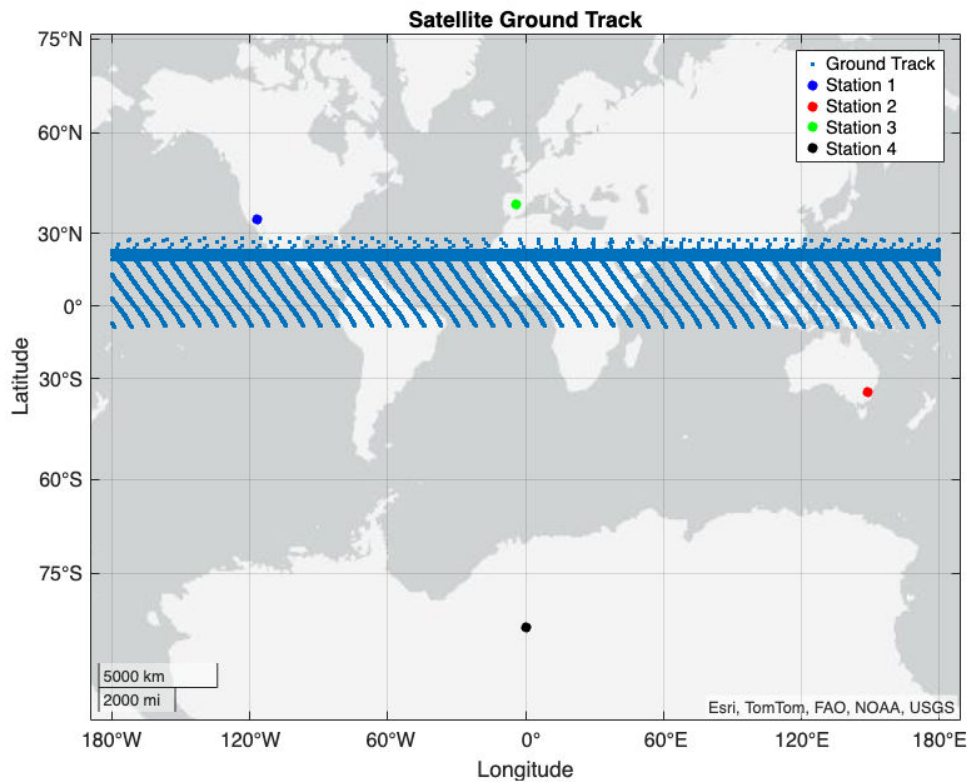
```
disp("Closest Approach of Lunar Trailblazer to the Moon is " +
min(flybyDistance) + " km at " + timeminflybyDist + " days.")
```

Closest Approach of Lunar Trailblazer to the Moon is 4081.4688 km at 250.5417 days.

Section 4: Plot Ground Track

```
% Convert Relative Position Vectors to EME
R_EM0toEME = inv(params.R_EM0toEM0);
XrelEarthEME = zeros(length(Ttotal),6);
for i = 1:length(Ttotal)
    XrelEarthEME(i,:) = [(R_EM0toEME*XrelEarth(i,1:3))',
(R_EM0toEME*XrelEarth(i,4:6))'];
end

plotGroundTrack(Ttotal,XrelEarthEME,params,st);
```



Section 5: Prossess Data and Get Converged deltaX0 and Pfinal

```
opts = struct();
opts.maxIter    = 11;
opts.earlyMode  = "drop";    % try also "downweight" to compare
opts.earlyDays  = 2;
opts.earlyScale = 10;

out = runBatchLSQ('0to14','ekf_k',opts);
```

runBatchLSQ: getting k_SRP from EKF (recommend 6to14)...

EKF 0to6: 5000 / 31930 (t=2.02 days)

EKF 0to6: 10000 / 31930 (t=2.94 days)

EKF 0to6: 15000 / 31930 (t=3.88 days)

EKF 0to6: 20000 / 31930 (t=4.46 days)

EKF 0to6: 25000 / 31930 (t=5.12 days)

EKF 0to6: 30000 / 31930 (t=5.70 days)

RR RMS prefit = 3.26954e-05 km/s (32.695 mm/s)

RR RMS postfit = 9.37663e-06 km/s (9.377 mm/s)

EKF k_SRP = 0.699682

Batch LSQ using ASTE583_Project_LTB_Measurements_0-6D_Truth.csv, ASTE583_Project_LTB_Measurements_6D-14D_

it	dX	cost	RR_RMS [mm/s]	Rho_RMS [m]
1	3.784e+03	1.479e+17	5530.362	7312674.892
2	3.204e+00	3.231e+08	0.988	341.723
3	1.496e-03	4.045e+05	0.542	10.947
4	5.817e-05	4.045e+05	0.542	10.947
5	4.238e-05	4.045e+05	0.542	10.947
6	2.611e-03	4.046e+05	0.542	10.947
7	2.579e-03	4.045e+05	0.542	10.947

8	6.025e-05	4.045e+05	0.542	10.947
9	4.648e-05	4.045e+05	0.542	10.947
10	1.602e-05	4.045e+05	0.542	10.947
11	1.983e-05	4.045e+05	0.542	10.947

Final Batch LSQ estimate at ET0 (mode=ekf_k):

x [km]	1.067657e+08	(± 3.847403e-01 3σ)
y [km]	1.148773e+08	(± 9.116523e-01 3σ)
z [km]	-3.258081e+04	(± 3.826600e+00 3σ)
vx [km/s]	-2.214337e+01	(± 2.109976e-07 3σ)
vy [km/s]	1.881438e+01	(± 9.304903e-08 3σ)
vz [km/s]	-9.259643e-02	(± 4.243887e-06 3σ)
k_SRP	6.996824e-01	(± 1.000000e+00 3σ)
lat_4 [rad]	-1.394519e+00	(± 8.875980e-07 3σ)
lon_4 [rad]	1.747701e-03	(± 2.250390e-06 3σ)
bias_rr [km/s]	1.630630e-06	(± 1.178839e-08 3σ)

```
deltaX0 = out.X_est - X0;           % correction to the a priori initial state
P_final = out.P_est;                 % batch covariance at ET0
```

```
disp('deltaX0 ='); disp(deltaX0.');
```

```
deltaX0 =
    1.0e+03 *
```

3.4280	1.5528	-0.4181	0.0000	0.0000	0.0000	-0.0003	0.0000	0.0000	0.0000
--------	--------	---------	--------	--------	--------	---------	--------	--------	--------

```
disp('P_final ='); disp(P_final);
```

```
P_final =
```

0.0164	0.0390	0.0208	-0.0000	-0.0000	-0.0000	0	-0.0000	-0.0000	0.0000
0.0390	0.0923	0.0471	-0.0000	-0.0000	-0.0000	0	-0.0000	-0.0000	0.0000
0.0208	0.0471	1.6270	-0.0000	-0.0000	-0.0000	0	-0.0000	0.0000	-0.0000
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	-0.0000
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.0000	0	0.0000	0.0000	0.0000
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.0000	0	0.0000	-0.0000	0.0000
0	0	0	0	0	0	0.1111	0	0	0
-0.0000	-0.0000	-0.0000	0.0000	0.0000	0.0000	0	0.0000	-0.0000	-0.0000
-0.0000	-0.0000	0.0000	0.0000	0.0000	-0.0000	0	-0.0000	0.0000	-0.0000
0.0000	0.0000	-0.0000	-0.0000	0.0000	0.0000	0	-0.0000	-0.0000	0.0000

Section 6: Maneuver Statistics

```
% Post Fit Intitial State
```

```
X0fit = X0 + deltaX0;           % From Filters in Section 5
X0fit(7) = 0.6996824;           % lock k_SRP to 0D value
```

```
% Post Fit Initial Covariance
```

```
Pfit = P_final;
```

```
% # of Monte Carlo Trials
```

```
ntrials = 1000;
```

```
% LCM Time Horizon
```

```
LCM_th = 30*params.tday;
```

```
% LCM STM Matrix
```

```

indexLCM      = floor((LCM-ET0)/step) + 1;
indexLCM_th   = indexLCM + LCM_th/step;
STM_LCM_Target = reshape(Xtotal(indexLCM_th,11:end),10,10) * ...
                    inv(reshape(Xtotal(indexLCM,11:end),10,10));

% Storage Vectors
deltaVLTmmag = zeros(ntrials,1);
deltaVLCmmag = zeros(ntrials,1);
posMagError  = zeros(ntrials,1);
velMagError  = zeros(ntrials,1);
Rerror       = zeros(ntrials,1);
Verror       = zeros(ntrials,1);

for i = 1:ntrials
    % ---- Step 1: sample initial state (but DO NOT perturb k_SRP) ----
    stateDeviation = randn(10,1).*sqrt(diag(Pfit));
    stateDeviation(7) = 0;
    X0samp          = X0fit + stateDeviation;

    % ---- Step 2: integrate to LTM ----
    % X0sampPreLTM = [X0samp; reshape(eye(10),100,1)];
    X0sampPreLTM = X0samp; % 10x1 only
    [TpreLTM,XpreLTM] = ode45(@projectDynamics, ET0:step:LTM, ...
                             X0sampPreLTM, options, params, sc);

    % ---- Step 4: sample LTM execution error ----
    deltaVLTM = [0.6931075; -0.8462091; 0.0956979] + randn(3,1)*0.005;
    deltaVLTmmag(i) = norm(deltaVLTM);

    % ---- Step 5: apply LTM and go to LCM ----
    X0sampPostLTM = XpreLTM(end,:)' ; % still 10x1
    X0sampPostLTM(4:6) = X0sampPostLTM(4:6) + deltaVLTM;
    [T_LTMtoLCM,X_LTMtoLCM] = ode45(@projectDynamics, LTM:step:LCM, ...
                                     X0sampPostLTM, options, params, sc);

    % ---- Step 6: error at LCM ----
    deltaXsampLCM = [X_LTMtoLCM(end,1:3) - Xtotal(indexLCM,1:3), ...
                    X_LTMtoLCM(end,4:6) - Xtotal(indexLCM,4:6)].';

    posMagError(i) = norm(deltaXsampLCM(1:3));
    velMagError(i) = norm(deltaXsampLCM(4:6));

    % ---- Step 6b: cleanup ΔV at LCM ----
    deltaVLCM = -inv(STM_LCM_Target(1:3,4:6))*STM_LCM_Target(1:3,1:3)* ...
                deltaXsampLCM(1:3) - deltaXsampLCM(4:6);
    deltaVLCmmag(i) = norm(deltaVLCM);

    % ---- Step 7: integrate to +30 days ----
    X0sampPostLCM = X_LTMtoLCM(end,:)' ;

```

```

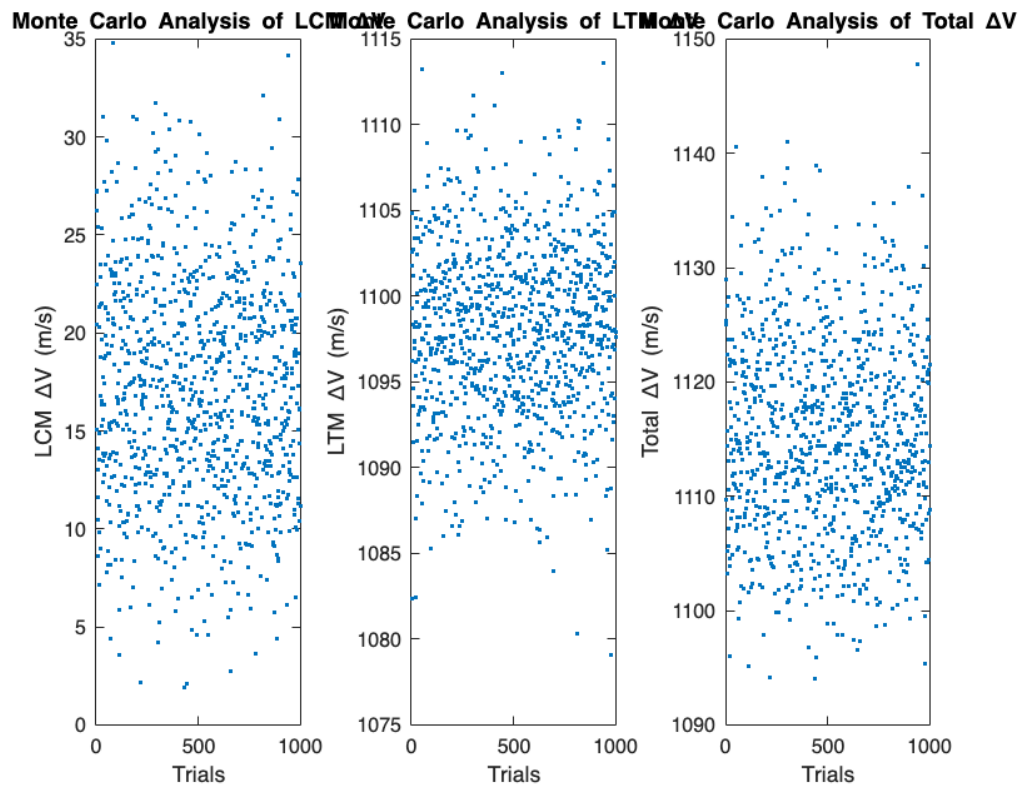
X0sampPostLCM(4:6) = X0sampPostLCM(4:6) + deltaVLCM;
[TpostLCM,XpostLCM] = ode45(@projectDynamics, ...
                             LCM:step:LCM+227*params.tday, ...
                             X0sampPostLCM, options, params, sc);

% ---- Step 9: errors at 30 days after LCM ----
Xsamp = [XpreLTM; X_LTMtoLCM(2:end,:); XpostLCM(2:end,:)];
Tsamp = [TpreLTM; T_LTMtoLCM(2:end); TpostLCM(2:end)];

Error(i) = norm(Xsamp(indexLCM_th,1:3) - Xtotal(indexLCM_th,1:3));
Verror(i) = norm(Xsamp(indexLCM_th,4:6) - Xtotal(indexLCM_th,4:6));
end

figure()
subplot(1,3,1)
plot(linspace(1,ntrials,ntrials),deltaVLCMmag*1000,'.')
xlabel("Trials")
ylabel("LCM \DeltaV (m/s)")
title("Monte Carlo Analysis of LCM \DeltaV")
subplot(1,3,2)
plot(linspace(1,ntrials,ntrials),deltaVLTMmag*1000,'.')
xlabel("Trials")
ylabel("LTM \DeltaV (m/s)")
title("Monte Carlo Analysis of LTM \DeltaV")
subplot(1,3,3)
plot(linspace(1,ntrials,ntrials),(deltaVLCMmag + deltaVLTMmag)*1000,'.')
xlabel("Trials")
ylabel("Total \DeltaV (m/s)")
title("Monte Carlo Analysis of Total \DeltaV")

```

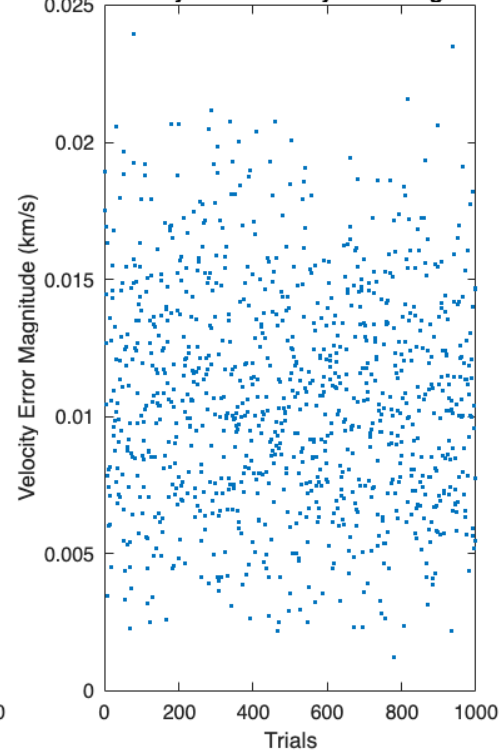
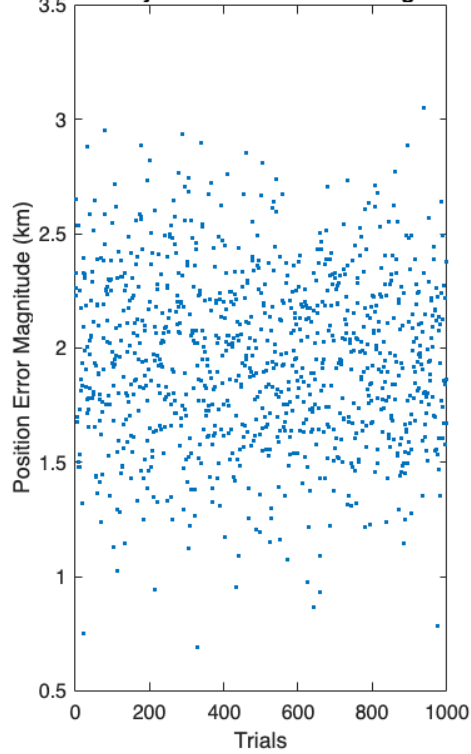


```
disp("The 99th Percentile for Total DeltaV is " + prctile(deltaVLCMmag +
deltaVLTMag, 99))
```

The 99th Percentile for Total DeltaV is 1.1361

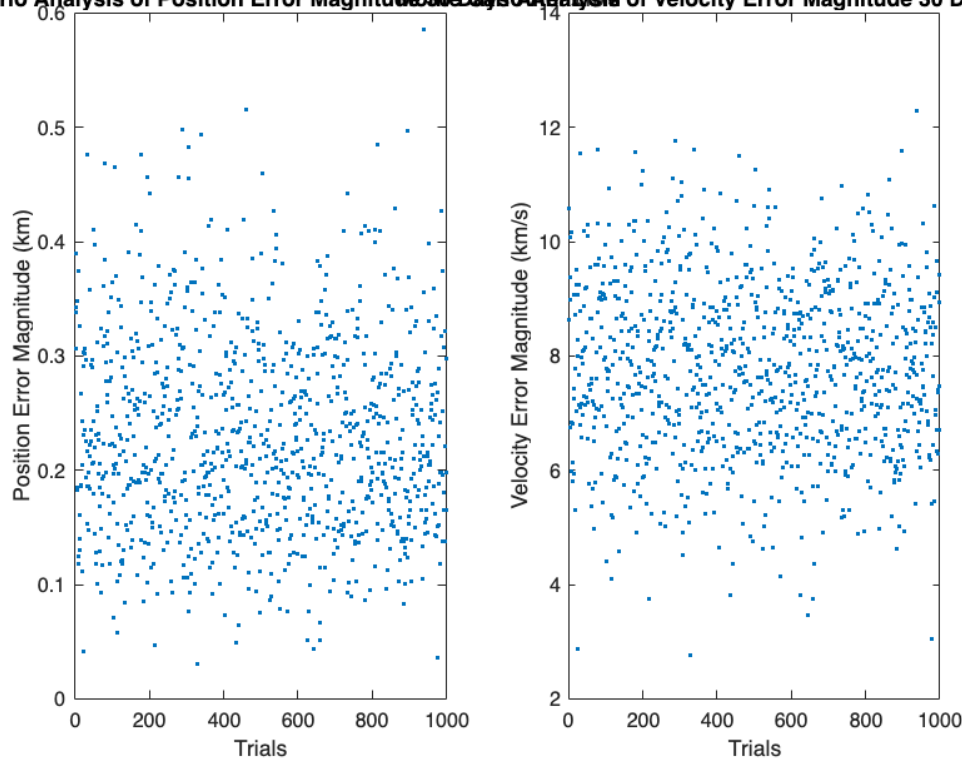
```
figure()
subplot(1,2,1)
plot(linspace(1,ntrials,ntrials),posMagError, '.')
xlabel("Trials")
ylabel("Position Error Magnitude (km)")
title("Monte Carlo Analysis of Position Error Magnitude at LCM")
subplot(1,2,2)
plot(linspace(1,ntrials,ntrials),velMagError, '.')
xlabel("Trials")
ylabel("Velocity Error Magnitude (km/s)")
title("Monte Carlo Analysis of Velocity Error Magnitude at LCM")
```

Figure 10: Monte Carlo Analysis of Position Error Magnitude at LCM



```
figure()
subplot(1,2,1)
plot(linspace(1,ntrials,ntrials),Rerror,'.')
xlabel("Trials")
ylabel("Position Error Magnitude (km)")
title("Monte Carlo Analysis of Position Error Magnitude 30 Days After LCM")
subplot(1,2,2)
plot(linspace(1,ntrials,ntrials),Verror,'.')
xlabel("Trials")
ylabel("Velocity Error Magnitude (km/s)")
title("Monte Carlo Analysis of Velocity Error Magnitude 30 Days After LCM")
```

Monte Carlo Analysis of Position Error Magnitude 30 Days After LCM and Monte Carlo Analysis of Velocity Error Magnitude 30 Days After LCM



```
disp("The Average for Postion Error 30 Days Post LCM is " + mean(Rerror))
```

The Average for Postion Error 30 Days Post LCM is 0.22988

```
disp("The 99th Percentile for Postion Error 30 Days Post LCM is " +  
prctile(Rerror, 99))
```

The 99th Percentile for Postion Error 30 Days Post LCM is 0.46686

```
disp("The Average for Velocity Error 30 Days Post LCM is " + mean(Verror))
```

The Average for Velocity Error 30 Days Post LCM is 0.0077773

```
disp("The 99th Percentile for Velocity Error 30 Days Post LCM is " +  
prctile(Verror, 99))
```

The 99th Percentile for Velocity Error 30 Days Post LCM is 0.011171

Section 7: Maneuver Execution

```
% --- Integration settings ---  
options = odeset('RelTol',1e-10,'AbsTol',1e-10);  
step     = 10;                % 10 s  
  
% --- Integrate OD solution to LTM (state only, 10x1) ---  
X0preLTM = X0fit;             % 10-state, no STM needed here  
[TpreLTM, XpreLTM] = ode45(@projectDynamics, ET0:step:LTM, ...
```

```

X0preLTM, options, params, sc);

% State at LTM
X0_LTM = XpreLTM(end,:).';

% --- NO-LTM trajectory: LTM -> LTM+1h ---
[Tpost_no, Xpost_no] = ode45(@projectDynamics, LTM:step:LTM+3600, ...
                             X0_LTM, options, params, sc);

% Combine ±1h around LTM
XnoLTM = [XpreLTM(end-360:end,:); Xpost_no(2:end,:)];
TnoLTM = [TpreLTM(end-360:end);    Tpost_no(2:end)];

% --- WITH-LTM trajectory: apply ΔV at LTM, then propagate ---
deltaVLTM = [0.6931075; -0.8462091; 0.0956979];

X0_LTM_burn      = X0_LTM;
X0_LTM_burn(4:6) = X0_LTM_burn(4:6) + deltaVLTM;

[Tpost_yes, Xpost_yes] = ode45(@projectDynamics, LTM:step:LTM+3600, ...
                             X0_LTM_burn, options, params, sc);

XwithLTM = [XpreLTM(end-360:end,:); Xpost_yes(2:end,:)];
TwithLTM = [TpreLTM(end-360:end);    Tpost_yes(2:end)];

% --- Convert to Earth-centered (same times for both) ---
xEarth = cspice_spkezr('Earth', TnoLTM.', 'ECLIPJ2000', 'NONE', 'Sun'); %
6xN
xEarth = xEarth.'; % N x 6

XrelnoLTM    = XnoLTM(:,1:6) - xEarth;
XrelwithLTM  = XwithLTM(:,1:6) - xEarth;

% --- Rotate to EME (vectorized) ---
R_EM0toEME = inv(params.R_EM0toEM0);

XrelnoLTMEME = XrelnoLTM;
XrelnoLTMEME(:,1:3) = (R_EM0toEME * XrelnoLTM(:,1:3).').';
XrelnoLTMEME(:,4:6) = (R_EM0toEME * XrelnoLTM(:,4:6).').';

XrelwithLTMEME = XrelwithLTM;
XrelwithLTMEME(:,1:3) = (R_EM0toEME * XrelwithLTM(:,1:3).').';
XrelwithLTMEME(:,4:6) = (R_EM0toEME * XrelwithLTM(:,4:6).').';

% --- Synthetic observations + noise ---
obsNoLTM = computeObservations(TnoLTM, XrelnoLTMEME, X0fit, params, st);
obsNoLTM(:,3) = obsNoLTM(:,3) + randn(size(obsNoLTM,1),1)*1e-3; % 1 m
obsNoLTM(:,4) = obsNoLTM(:,4) + randn(size(obsNoLTM,1),1)*1e-7; % 1 mm/s

```

```

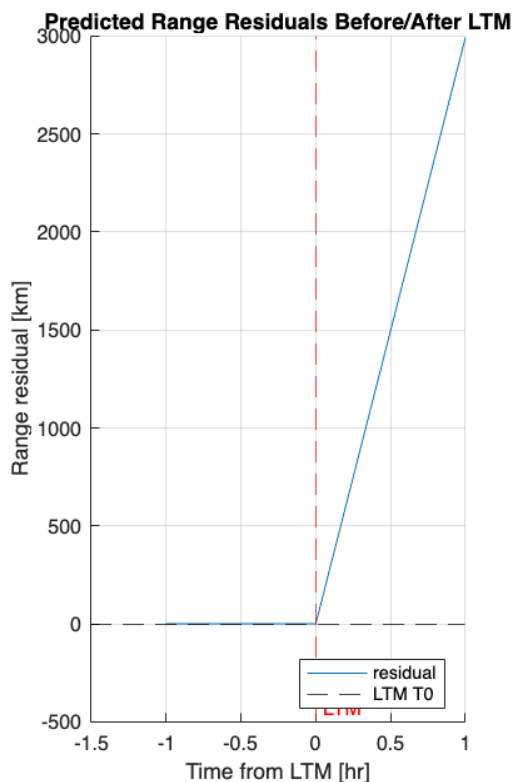
obsWithLTM = computeObservations(TwithLTM, XrelwithLTMEME, X0fit, params,
st);
obsWithLTM(:,3) = obsWithLTM(:,3) + randn(size(obsWithLTM,1),1)*1e-3;
obsWithLTM(:,4) = obsWithLTM(:,4) + randn(size(obsWithLTM,1),1)*1e-7;

residuals = obsWithLTM - obsNoLTM;

% --- Plots ---
t_hr = (TnoLTM - LTM) / 3600;

figure;
subplot(1,2,1); hold on; grid on;
plot(t_hr, residuals(:,3));
yline(0,'k--');
xline(0,'r--','LTM','LabelOrientation','horizontal','LabelVerticalAlignment',
'bottom');
title('Predicted Range Residuals Before/After LTM');
xlabel('Time from LTM [hr]');
ylabel('Range residual [km]');
legend('residual','LTM T0','Location','best');

```



```

% subplot
% (1,2,2); hold on; grid on;
%
% plot(t_hr, residuals(:,4));

```

```

% yline(0,'k--');
%
xline(0,'r--','LTM','LabelOrientation','horizontal','LabelVerticalAlignment'
,'bottom');
% title('Predicted Range-rate Residuals Before/After LTM');
% xlabel('Time from LTM [hr]');
% ylabel('Range-rate residual [km/s]');
% legend('residual','LTM T0','Location','best');

%% === Export everything for Python / post-processing ===

outFile = 'LTB_maneuver_all.mat';

save(outFile, ...
    ... % OD-ish info
    'X0', 'deltaX0', 'X0fit', 'Pfit', ...
    ... % reference trajectory (Sun/Earth centered) if you have it in this
script
    'Ttotal', 'Xtotal', ... % sc wrt Sun, full duration
    ... % LTM +/- 1 hour with and without burn
    'TnoLTM', 'XnoLTM', ...
    'TwithLTM', 'XwithLTM', ...
    'XrelnoLTMEME', 'XrelwithLTMEME', ...
    ... % synthetic measurements around LTM
    'obsNoLTM', 'obsWithLTM', 'residuals', ...
    ... % Monte Carlo statistics at LCM and +30 days
    'deltaVLTmmag', 'deltaVLCMmag', ...
    'posMagError', 'velMagError', ...
    'Rerror', 'Verror', ...
    '-v7.3');

fprintf('Saved maneuver + MC data to %s\n', outFile);

```

Saved maneuver + MC data to LTB_maneuver_all.mat

```

% LTM residuals time series
T_LTM = table();
T_LTM.t_days = (TnoLTM - LTM)/86400;
T_LTM.range_res = residuals(:,3); % km
T_LTM.rr_res = residuals(:,4); % km/s
writetable(T_LTM, 'LTM_residuals.csv');

% Monte Carlo summary
T_MC = table();
T_MC.deltaVLTmmps = deltaVLTmmag*1000; % m/s
T_MC.deltaVLCMmps = deltaVLCMmag*1000; % m/s
T_MC.totalDVmps = (deltaVLTmmag + deltaVLCMmag)*1000;
T_MC.R_LCM_km = posMagError;

```

```
T_MC.V_LCM_kmps      = velMagError;  
T_MC.R_30d_km        = Rerror;  
T_MC.V_30d_kmps      = Verror;  
writetable(T_MC, 'MC_LCM_summary.csv');  
  
fprintf('Also wrote LTM_residuals.csv and MC_LCM_summary.csv\n');
```

Also wrote LTM_residuals.csv and MC_LCM_summary.csv